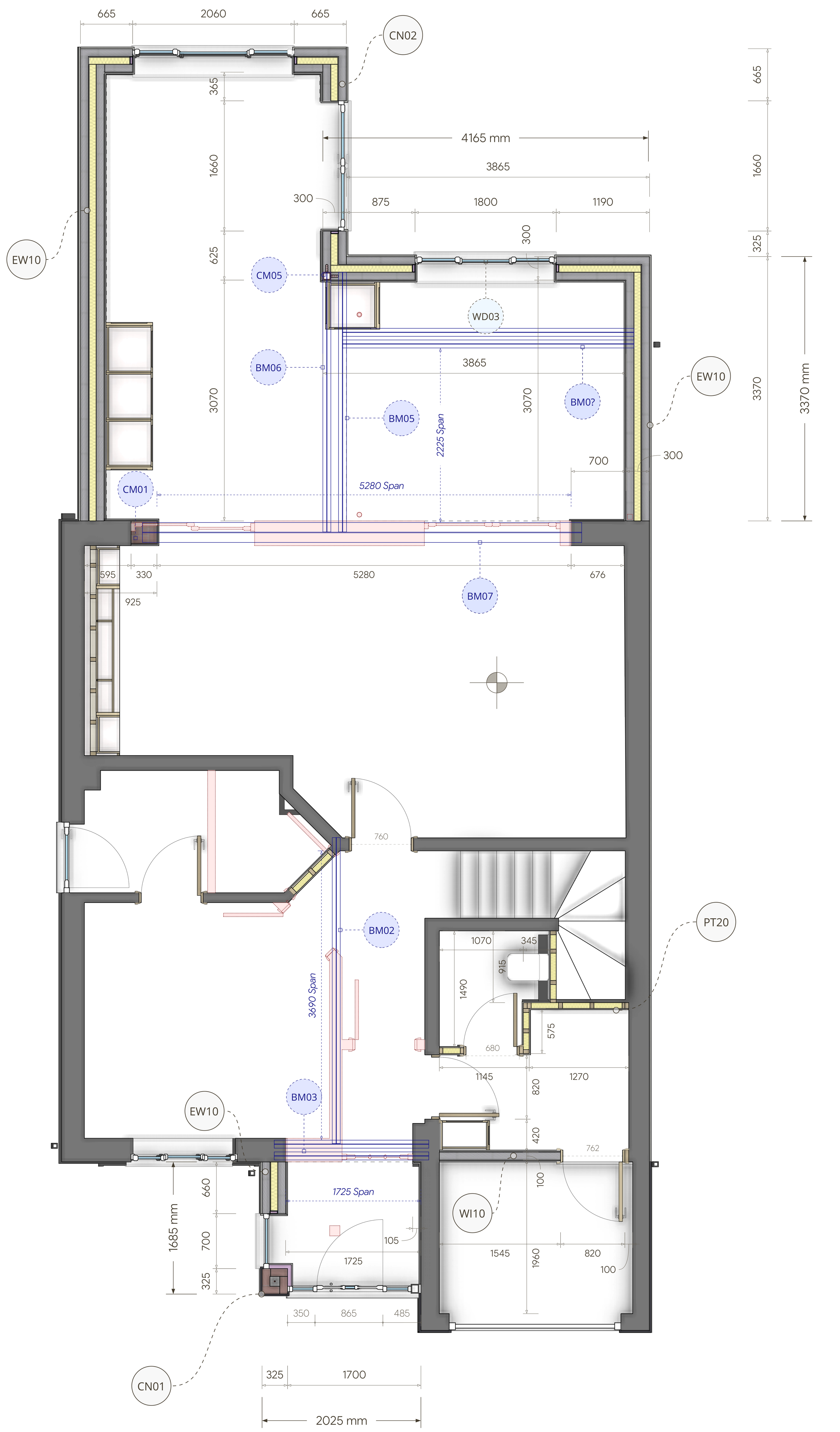
EW10	Cavity Wall | PIR Insulation & Rendered Finish
1. Internal Finish	
1.1. Plaster Skim	**3mm** Gypsum-based skim coat
1.2. Plasterboard	**12.5mm** Standard plasterboard
1.3. Fixing & Finish	Boards fixed with adhesive or mechanical fixings as per manufacturer's guidance. Apply a **3mm skim** coat to finish. Ensure boards are tightly butted, with all joints taped and sealed before plastering.
1.4. Consideration	Consider moisture-resistant boards if required.
1.5. Accreditations	CE marked
1.6. Standards	**BS 8212** | Internal plastering & drylining
2. Inner Leaf Blockwork	
2.1. Material	**Celcon Standard 3.6N** Aerated Concrete Block
2.2. Thickness	**100mm**
2.3. Mortar	1:6 Cement : Sand mix. Ensure adequate workability and compressive strength. Add frost-resistant additives in cold conditions if required.
2.4. Consideration	Pressure-treated timber packers or DPC to be provided as per detail at junctions.
2.5. Accreditations	CE marked blocks
2.6. Standards	**BS EN 771-3** | Masonry units, **BS 5628** | Structural use of masonry
3. Insulation (Partial Fill)	
3.1. Thickness	**90mm** PIR Insulation Board
3.2. Material	Recticel Eurowall+ (or equivalent high-performance PIR)
3.3. Critical Info	Ensure boards are tightly butted. Install as per manufacturer's instructions. Use foil tape at joints with a minimum 50mm overlap to ensure a continuous secondary vapour barrier.
3.4. Accreditations	CE marked & BBA certified
3.5. Standards	**BS EN 13165:2012+A2:2016** | Thermal insulation products
4. Unventilated Air Void	
4.1. Thickness	**10mm** Minimum
4.2. Critical Info	Maintain a clear cavity space. Ensure no debris or mortar snots bridge the gap. The cavity must be kept free of insulation off-cuts or obstructions to prevent thermal bridging.
5. Cavity Wall Ties	
5.1. Material	**Stainless Steel** cavity wall ties
5.2. Product	Ancon Staifix HRT4 or equivalent
5.3. Important	Spacing at **900mm c/c horizontally** and **450mm vertically**. Additional ties around openings at **225mm vertical centres** and within **300mm** of reveals.
5.4. Note	Ties must slope slightly towards the external leaf to shed water outwards. Embed minimum **50mm** into each masonry leaf.
5.5. Accreditations	CE marked & BBA certified
5.6. Standards	**BS EN 845-1** | Masonry wall ties and kits
6. Outer Leaf Blockwork	
6.1. Material	**Celcon Standard 3.6N** Aerated Concrete Block
6.2. Thickness	**100mm**
6.3. Mortar	1:5 Cement : Sand mix, similar to inner leaf. Sulphate-resistant cement if ground conditions necessitate.
6.4. Consideration	Ensure proper alignment and bonding pattern. Avoid bridging cavity with mortar droppings.
6.5. Accreditations	CE marked blocks
6.6. Standards	**BS EN 771-3** | Masonry units, **BS 5628** | Structural use of masonry
7. External Finish (Render)	
7.1. Material	Silicon-based external render system, approx. **15mm** total thickness
7.2. Product	Weber Monocouche or a similar performing product such as K-Rend etc
7.2. Critical Info	Apply as per manufacturer's guidelines. Ensure adequate key and curing.
7.3. Consideration	Silicon-based renders improve water repellence while allowing vapour permeability.
7.4. Standards	**BS EN 13914-1** | Design, preparation and application of external rendering
U-Value & Part L Compliance *(In accordance with BS EN ISO 6946)*	
8.1. Required U-Value	**0.180 W/m²K** | Min for Extension Walls **Approved Part L** *2022 Updates*
8.2. Achieved U-Value	0.174 W/m²K - ✓ Achieved ✓ - Exceeds Part L requirement

Global Height Benchmark

Datum GCP | Ground Floor Finished Floor Level

All height measurements on this drawing are taken from the existing ground floor finished floor level, which serves as the universal site datum benchmark unless otherwise specified. Heights are set globally based on the primary ground control point established during the survey. If exact coordinates of this ground control point are required, please contact Noble Architecture.



PL10 | Technical Plan - Ground Floor

Printed Scale Factor | 1mm : 50mm

RE20	Masonry Restraint – Wall Starter Kits
4.1. Material	**Stainless Steel**; guarantees corrosion resistance and long-term durability.
4.2. Product	**Ancon Staifix Universal Wall Starter System** or equivalent system.
4.3. Dimensions	Standard kit length of **2.4 meters**; adjustable to fit varying wall heights.
4.4. Installation	At perpendicular wall abutments where new walls meet the host building, secure wall starter kits with corrosion-resistant fixings spaced spaced no more than **600mm** apart vertically.
4.5. Installation	Install **vertical DPC** at all abutments to prevent moisture ingress, particularly where solid walls without a drainage cavity are present, as this could compromise damp-proofing at the abutments.
4.6. Critical Note	Wall starter kit ties should be securely embedded in the mortar bed joints, with all ties spaced vertically according to the manufacturer's instructions.
4.7. Rationale	Wall starter kits tie new masonry walls to existing structures to prevent movement or separation.
4.8. Accreditations	**CE marked** and **BBA certified** to ensure quality and performance.
4.9. Standards	**BS EN 845-1:2013+A1:2016** | Masonry wall ties and kits **BS EN 1996-1-1:2005 (Eurocode 6)** | Masonry structures

RE10	Cavity Wall Ties
2.1. Material	**Stainless Steel**; guarantees corrosion resistance and long-term durability.
2.2. Product	**Ancon Staifix HRT4** or equivalent.
2.3. Important	Wall ties must be spaced at **900mm c/c horizontally** along the wall and **450mm vertically**.
2.4. Important	Additional wall ties must be installed around openings, reducing the spacing to **225mm vertically** and ensuring they are located within **300mm** of the opening reveal.
2.5. Note	Ties must be embedded to a minimum depth of **50mm** into each masonry leaf to ensure adequate bonding between the leaves.
2.6. Installation	Wall ties should slope slightly downwards towards the external leaf, rather than upwards, this is to ensure any water breaching the cavity drains out, preventing moisture ingress.
2.8. Accreditation	**CE marked** and **BBA certified** to ensure quality and performance.
2.9. Standards	**BS EN 845-1** for wall ties **BS 5628** for the structural use of masonry in buildings.

CN01	Insulated Solid Brick Corner Column
1. Column Construction	
1.1. Material	**Class B Engineering Brick** – Min. 75 N/mm² Compressive Strength
1.2. Mortar	**M6 Grade** – 1:4 Cement : Sand Mix
1.3. Dimensions	325mm x 325mm External Size
2. Concrete Core	
2.1. Dimensions	125mm x 125mm Solid In-Situ Concrete Core
2.2. Concrete Strength	**C25/30 Mix** – 25 N/mm² Cylinder Strength
2.3. Reinforcement	**1No. 12mm High Yield Rebar** Vertically Placed – Lapped and Anchored Per BS EN 1992-1-1
2.4. Ties	6mm Horizontal Reinforcing Ties @ min h450mm
2.5. British Standards	Comply with **BS EN 1992-1-1:2004** | Eurocode 2
3. Internal Insulated Lining	
3.1. Insulated Board	**Phenolic Foam-Backed Plasterboard** – e.g., Kingspan Kooltherm K118
3.2. Thickness	**62.5mm Total** – 50mm Insulation + 12.5mm Plasterboard
3.3. Thermal Conductivity	**λ ≤ 0.020 W/mK**
3.4. Fixing Method	**Dot & Dab** Adhesive with Full Perimeter Bead + Vertical Dabs
3.5. Accredited Products	CE Marked & **BBA Certified**
4. Important Notes	
4.1. Request For Revision	- Phone Call received on **20th March 2025** requesting amendment - Replaced original AMSD steel wind post design - New Masonry Column **Must** be approved by BCO
4.2. Approval Of Detail	Must be **agreed with the Building Inspector** on site prior to execution
4.3. Liability Omission	This alteration is made **outside the original engineer's design liability** and must be structurally validated on site.

ST01	Structural Engineering Notes
1. Scope	Coordination of structural elements and integration within the ceiling.
2. Structural Members	- All beams illustrated on drawings are for **coordination purposes only**. - Final beam sizes and specifications to be provided by the Structural Engineer.
3. Concealment	- All beams must be **fully concealed** within the ceiling structure.